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Matching engines and graphical user interfaces for partner recommendations and lead sharing

Abstract

Systems and methods for assisting in pairing entities in an electronic payments processing network are disclosed herein. For example, systems and methods are disclosed herein for generating a customized user interface for displaying a plurality of potential partners/resellers for pairing with one or more leads. Systems and methods herein may be used to receive the identity of one or more potential partners/resellers having a plurality of attributes, receive the identity of one or more new leads and a plurality of lead attributes, select one or more potential partners/resellers having attributes that are compatible with the plurality of lead attributes, and display the one or more potential partners/resellers on a graphical user interface.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7251624	12/2006	Lee	705/26.1	G06Q 30/06
2003/0171975	12/2002	Kirshenbaum	705/7.33	G06Q 30/0204
2008/0228824	12/2007	Kenedy	N/A	G16B 20/40
2013/0191223	12/2012	Harris et al.	N/A	N/A
2013/0246120	12/2012	Chang	705/7.29	G06Q 30/0201
2015/0134488	12/2014	MacLaurin	705/26.81	G06Q 30/0625
2016/0055600	12/2015	Chau	705/15	G06Q 30/0206
2016/0110674	12/2015	Berman	705/7.14	G06Q 10/063112

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2009011915	12/2008	WO	G06Q 10/0635

OTHER PUBLICATIONS

Sailesh Venkatraman, Business Partnering When Does It Work?, Strategic Finance, pp. 47-53, Feb. 2015. (Year: 2015). cited by examiner

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This patent application is a continuation of and claims the benefit of priority to U.S. application Ser. No. 17/193,585, filed on Mar. 5, 2021, which is a continuation of and claims the benefit of priority to U.S. application Ser. No. 15/367,944, now U.S. Pat. No. 10,970,763, filed on Dec. 2, 2016, the entireties of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The embodiments disclosed herein relate generally to matching engines and graphical user interfaces (“GUIs”) for data processing networks. More specifically, the embodiments disclosed herein relate to matching engines and GUIs for partner recommendations and lead sharing relating to payments processing networks, and systems and methods thereof.

BACKGROUND

(2) Many brick and mortar merchants use electronic point of sale (POS) terminal systems to perform transaction accounting and payment processing, including both for payment cards and electronic payments. One development in POS terminal systems has been the establishment of third party electronic payment management service providers (also known as acquiring banks and their affiliates) who securely and reliably aggregate and direct electronic payment requests from merchants' POS terminals through payment networks to payment card companies' payment systems, and thus act as payment intermediaries. Such electronic payment management services may also provide merchants greater control over the processing of electronic payments via Arbitrated Enhanced Payment (AEP) services including, but not limited to, how rapidly a given payment is processed, what entity processes a given payment, and/or enhanced payment security measures. From time to time, a given merchant may select a new electronic payment management service. The process of integrating a new merchant account's POS terminal system(s) and corresponding payments stream into an arbitrated enhanced payments processing system (AEPPS) is commonly termed “merchant boarding.”

(3) Merchants are now often assisted with merchant boarding by a variety of intermediary partners (sometimes known as resellers) including, but not limited to, independent sales organizations (ISOs) and POS system developers and vendors. Such partners may also assist merchants with aspects of AEP services beyond merchant boarding. For example, a merchant's POS terminal's components may be sourced from several manufacturers, and may be integrated and maintained over time by third party POS system developers and vendors. An ISO may, for example, serve as an aggregating entity that may modify or augment transactional content exchanged with a given POS terminal system.

(4) To maximize earning potential, acquiring banks may want to add as many new merchants (sometimes referred to by acquiring banks and partners/resellers as “leads”) to an electronic payment management service as possible. Leads may be recruited to an electronic payment management service through initial contact with an intermediary partner such as an ISO or a POS system vendor, or leads may also directly contact electronic payment management services, e.g., via signing up online without the involvement of a partner. Acquiring banks and/or partners may also wish to share lead information with one another to, for example, expedite and streamline the merchant boarding process and the providing of AEP services to a lead. There exists a need for matching engines and graphical user interfaces for partner recommendations and lead sharing, and systems and methods thereof.

(5) There further exists a need for a system and method to track and promote assistance provided by partners to leads during the merchant boarding process and with AEP services beyond merchant boarding, such as implementing enhanced payment security measures for a lead. There also exists a need for a GUI suitable for providing a user with representations and summaries of the assistance provided by partners to leads during the merchant boarding process and with AEP services.

SUMMARY

(6) Methods are disclosed for automatically generating a customized display on a graphical user interface by: receiving, at a processor, an electronic database comprising a plurality of potential partners and a plurality of attribute categories, each potential partner having an attribute corresponding to each attribute category; receiving, at the processor, a plurality of threshold eligibility attributes, each threshold eligibility attribute corresponding to at least one of the plurality of attribute categories; using the processor to automatically narrow the plurality of potential partners using one or more of the plurality of threshold eligibility attributes as one or more narrowing attributes; using the processor to construct an electronic questionnaire, wherein the questionnaire comprises a plurality of answerable prompts that may be displayed on a graphical user interface; using the processor to transmit the questionnaire to an interactive lead user interface over an electronic network; receiving, at the processor, one or more answers to the questionnaire via the lead user interface over the electronic network, wherein each answer comprises at least one lead attribute that corresponds to one of the plurality of attribute categories; using the processor to automatically narrow the plurality of potential partners using the at least one lead attribute as at least one narrowing attribute; and displaying the at least one lead attribute and the narrowed plurality of potential partners on a results user interface of the graphical user interface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments and together with the description, serve to explain the principles of the disclosure.
- (2) FIG. 1A is a schematic diagram of a system depicting the general arrangement between merchants, partners, consumers, electronic payment management service providers, and payment networks, according to an exemplary embodiment.
- (3) FIG. 1B is a schematic diagram of a system depicting an arrangement between a processor, a matching engine, and a database, according to an exemplary embodiment.
- (4) FIG. 2 shows an exemplary view of a graphical user interface dashboard, according to an exemplary embodiment.
- (5) FIG. 3 shows an exemplary view of a graphical user interface providing partner management tools, according to an exemplary embodiment.
- (6) FIG. 4 shows an exemplary view of a graphical user interface providing partner details, according to an exemplary embodiment.
- (7) FIG. 5 shows another exemplary view of a graphical user interface providing further partner details, according to an exemplary embodiment.
- (8) FIG. 6 shows another exemplary view of a graphical user interface providing further partner details, according to an exemplary embodiment.
- (9) FIG. 7 shows an exemplary view of a graphical user interface providing information about leads, according to an exemplary embodiment.
- (10) FIG. 8 shows an exemplary view of a graphical user interface providing lead details, according to an exemplary embodiment.
- (11) FIG. 9 shows another exemplary view of a graphical user interface providing further lead details, according to an exemplary embodiment.
- (12) FIG. 10 shows an exemplary view of a graphical user interface providing a lead questionnaire, according to an exemplary embodiment.
- (13) FIG. 11 shows another exemplary view of a graphical user interface providing a lead questionnaire, according to an exemplary embodiment.
- (14) FIG. 12 shows a graphic representation of an algorithm design for narrowing potential partners

according to an exemplary embodiment.

(15) FIG. 13 shows an exemplary view of a graphical user interface providing partner performance details, according to an exemplary embodiment.

(16) FIG. 14 shows an exemplary view of a graphical user interface providing a summary of partner performance, according to an exemplary embodiment.

(17) FIG. 15 shows an exemplary view of a graphical user interface providing user information, according to an exemplary embodiment.

(18) FIG. 16 is a block diagram of a method for generating and presenting a graphical user interface showing a subset of a plurality of potential partners for a given lead, according to an exemplary embodiment.

(19) FIG. 17 is a block diagram of a method for assisting in generating and presenting a graphical user interface showing a subset of a plurality of potential partners for a given lead, according to an exemplary embodiment.

DESCRIPTION OF THE EMBODIMENTS

(20) Reference will now be made in detail to exemplary embodiments, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

(21) FIG. 1A depicts a payments processing network **100** generally comprising a plurality of merchants **112a**, **112b**, new leads **114a**, **114b**, partner/resellers **116a**, **116b**, **116c**, acquiring banks **102a**, **102b**, issuing banks **106a**, **106b**, and consumers **108a**, **108b**, all provided in communication with each other via one or more payment networks **104a**, **104b**. The components of processing network **100** may be connected by any combination of wired or wireless networks, including, for example, PSTNs and/or the Internet. As shown in FIG. 1, acquiring bank **102a** may have partnerships with one or more payment networks **104a**, **104b**, such that the acquiring bank **102a** may process payments through, and on behalf of, payment networks **104a**, **104b**. Payment networks **104a**, **104b** may in turn have partnerships with one or more issuing banks **106a**, **106b**. Issuing banks **106a**, **106b** may hold accounts for one or more consumers **108a**, **108b** with accounts at issuing banks **106a**, **106b**. Consumers **108a**, **108b** may have payment vehicles, e.g., one or more payment cards **110a**, **110b** (e.g., credit cards, debit cards, stored value cards, etc.), each of which may be affiliated with a payment network, e.g., payment network **104a** or **104b**. Consumers **108a**, **108b** may be able to use their payment cards **110a**, **110b** at one or more merchants **112a**, **112b** and/or one or more new merchant leads **114a**, **114b** to purchase goods and services. Merchants **112a**, **112b** may already be using the services of an acquiring bank such as acquiring bank **102a** through, for example, at least one partner/reseller (or “partner”) **116a**, **116b**, whereas new merchant leads **114a**, **114b** may be potential new customers of acquiring bank **102a** and/or a partner such as partner **116b** or **116c**. Partners **116a**, **116b** may act to provide merchant boarding services and Arbitrated Enhanced Payment (AEP) services to merchants **112a**, **112b**, and may act as intermediaries between acquiring bank **102a** and merchants **112a**, **112b**. New merchant leads **114a**, **114b** not yet affiliated with a partner or acquiring bank **102a** may have contacted acquiring bank **102a** and expressed interest in using electronic payment processing services provided by acquiring bank **102a** and the payment networks **104a**, **104b**. Partners **116b**, **116c** may be candidates for pairing with, and providing services (such as merchant boarding and AEP services) to, new leads **114a**, **114b**.

(22) Acquiring banks **102a**, **102b** may be entities that provide a variety of electronic payment processing services to merchants **112a**, **112b**, with the assistance of one or more intermediary partners **116a**, **116b**, **116c**. For example, acquiring bank **102a** may be an entity that receives payment information from a transaction which occurs at a POS terminal at, for example, merchant **112a**. The payment information may be, for example, payment card information encoded in the magnetic stripe or EMV chip of payment card **110a** and a payment amount of a transaction being made by, for example, consumer **108a** with merchant **112a** using the payment card account

associated with payment card **110a**. Acquiring bank **102a** may process the information, and may send the information to the consumer's respective issuing bank, e.g., issuing bank **106a**, via an appropriate payment network **104a**, depending on the particulars of payment card **110a**. Processing the information may include, for example, determining the identities of payment network **104a** and issuing bank **106a** associated with the particular payment card **110a**.

(23) Acquiring banks **102a**, **102b** may also receive information from payment networks **104a**, **104b**, such as a confirmation or rejection of an attempted transaction using payment cards **110a**, **110b**, and may convey that information to the appropriate POS terminal. Moreover, acquiring banks **102a**, **102b** may provide security and/or encryption services to merchants **112a**, **112b** and payment networks **104a**, **104b**, such that payments processed at one or more POS terminals of merchants **112a**, **112b** may be completed with a decreased risk of data or financial theft or loss. Acquiring banks **102a**, **102b** may be located, for example, remotely from merchants **112a**, **112b** that use their services, and may, for example, interact with merchants **112a**, **112b** primarily over an electronic network, such as a data network or the Internet. Acquiring banks **102a**, **102b** may also interact with one or more partners **116a**, **116b** that provide specific services to merchants **112a**, **112b**, as described above. For example, if a POS terminal that is initiating a given transaction was provided to merchant **112a** by partner **116a**, then partner **116a** may perform some additional processing of the transaction before or as the transaction is forwarded to acquiring bank **102a**.

(24) Payment networks **104a**, **104b** may be, for example, networks that relay debit and/or credit transactions to and from various accounts at issuing banks **106a**, **106b**. For example, payment network **104a** may have a partnership program with issuing bank **106a**, through which issuing bank **106a** may provide a payment card account to consumer **108a** associated with payment network **104a**. Payment networks **104a**, **104b** may also be partnered with one or more acquiring banks **102a**, **102b**, which may manage payment transactions associated with payment networks **104a**, **104b**. Examples of payment network brands include, e.g., Visa, MasterCard, Discover, and American Express. Multiple payment networks **104a**, **104b** may be partnered with a single acquiring bank, such as acquiring bank **102a**.

(25) Issuing banks **106a**, **106b** may be banks that manage payment accounts associated with one or more of payment networks **104a**, **104b**, on behalf of one or more of consumers **108a**, **108b**. For example, issuing bank **106a** may allow for consumer **108a** to build up a revolving credit balance at issuing bank **106a**, and may periodically receive payments from consumer **108a** to pay down the balance. Consumers **108a**, **108b** may be individuals, companies, or other entities having accounts with one or more of issuing banks **106a**, **106b** affiliated with one or more of payment networks **104a**, **104b**. Each of consumers **108a**, **108b** may generally have at least one payment card such as payment card **110a** or payment card **110b** associated with each payment account held by that consumer. Each of consumers **108a**, **108b** may have multiple accounts with multiple issuing banks **106a**, **106b**, which may be affiliated with the same or different payment networks **104a**, **104b**.

(26) Merchants **112a**, **112b** may be merchants offering goods and/or services for sale to consumers **108a**, **108b**, who have contracted with an acquiring bank, such as acquiring bank **102a**, optionally with one or more of intermediary partners **116a**, **116b**. Merchants **112a**, **112b** may have been equipped with POS terminals (e.g., provided by one or more of partners **116a**, **116b**) configured to receive payment information from payment cards such as payment cards **110a**, **110b** and relaying received payment information to acquiring bank **102a**.

(27) New leads **114a**, **114b** may be merchants offering (or preparing to offer) goods and/or services for sale to consumers such as consumers **108a**, **108b**, who do not yet use the services of an acquiring bank such as acquiring bank **102a**, and are not yet paired with a partner such as partners **116b**, **116c**, but who may be interested, or potentially interested, in doing so. New leads **114a**, **114b** may have expressed such interest in the services of acquiring bank **102a** by, for example, contacting one of partners **116b**, **116c**, or contacting acquiring bank **102a**, e.g., through a web form provided on the website of acquiring bank **102a**. In cases where a new lead, such as new lead **114a**,

has expressed interest in using the services of an acquiring bank such as acquiring bank **102a** and has contacted acquiring bank **102a** directly, acquiring bank **102a** may desire to pair the new lead **114a** with an intermediary partner such as partner **116b** who can assist the new lead **114a** with providing POS terminals and related services, merchant boarding, AEP services, and/or other services needed for the new lead **114** to integrate the services of acquiring bank **102a** into its business.

(28) Partners **116a**, **116b**, **116c** may be, for example, independent sales organizations (ISOs), or POS system developers or vendors that provide electronic payments processing services directly to one or more merchants such as merchants **112a**, **112b** and/or new leads such as new leads **114a**, **114b**. For example, partners **116a**, **116b**, **116c** may be ISOs that may sell credit card security services, gift card services, or other “value-added” payments-related goods and services offered by acquiring bank **102a** to one or more of merchants **112a**, **112b** and/or new leads **114a**, **114b**. As a further example, partners **116a**, **116b**, **116c** may be POS vendors that may sell or lease POS terminals to one or more of merchants **112a**, **112b** and/or new leads **114a**, **114b**, and/or may maintain such POS terminals for the one or more of merchants **112a**, **112b** and/or new leads **114a**, **114b**. POS terminals may be made by multiple third-party manufacturers, and POS vendors may additionally serve as intermediaries between merchants **112a**, **112b** and/or new leads **114a**, **114b** and third-party POS terminal manufacturers. In one embodiment, partners **116a**, **116b**, **116c** who provide software to one or more of merchants **112a**, **112b** and/or new leads **114a**, **114b** may be referred to as independent software vendors or “ISVs.”

(29) FIG. 1B depicts an exemplary view of an arrangement of systems in an acquiring bank system of acquiring bank **102a**, for enabling acquiring bank **102a** to evaluate partners **116a**, **116b**, **116c** and match partners **116a**, **116b**, **116c** with one or more of new leads **114a**, **114b**. As shown in FIG. 1B, in general, the system of acquiring bank **102A** may comprise a processor **150**, database **154**, and matching engine **156**. Processor **150** may be configured to send and/or receive data from one or more external sources, such as a computer **152**. Processor **150** may be connected to at least one database **154**, which may store data related to one or more of partners **116a**, **116b**, **116c** and/or new leads **114a**, **114b**. Both processor **150** and database **154** may be further connected to matching engine **156**, which may execute one or more matching algorithms to pair one or more of partners **116a**, **116b**, **116c** with one or more of new leads **114a**, **114b**.

(30) Processor **150** may be a computer, such as a server computer, configured to send, receive and process data transmitted between one or more computers, such as computer **152**, one or more databases, such as database **154**, and matching engine **156**. Processor **150** may be connected to computer **152**, database **154**, and matching engine **156** over one or more of a wired or wireless electronic network, such as a local area network or a wide area network (e.g., the Internet). For example, processor **150** may be connected to computer **152** over the Internet, and may be connected to database **154** and matching engine **156** over a local area network.

(31) In some embodiments, processor **150** may be a computer configured to generate graphical user interfaces, lists, database files, or other types of electronic files and transmit them to, for example, computer **152**. For example, processor **150** may be configured to receive data regarding one or more new leads **114a**, **114b** from computer **152**, automatically process received data into, for example, an electronic database format, and automatically transmit the processed received data to database **154** and/or matching engine **156**. As a further example, processor **150** may be configured to receive one or more narrowing attributes from, for example, computer **152**, and transmit the received narrowing attributes to the matching engine **156**. Processor **150** may also be configured to receive data from database **154** and/or matching engine **156** and generate a configuration of received data to transmit to, for example, computer **152**. The configuration of received data may be, for example, a database file, a text file, a graphical user interface display, or a combination thereof. For example, processor **150** may be configured to receive a list of potential partners **116a**, **116b**, **116c** from matching engine **156**, generate a graphical user interface showing the received list,

and transmit the generated graphical user interface to, for example, computer **152**. As a further example, processor **150** may be configured to receive a list of potential partners **116a**, **116b**, **116c** from matching engine **156**, generate a database file including the received list and additional data about potential partners **116a**, **116b**, **116c**, and transmit the database file to, for example, computer **152**. In some embodiments, processor **150** may be configured to transmit a configuration of received data to database **154** or matching engine **156**.

(32) Computer **152** may be a computer configured to send data to, and receive data from, processor **150**. For example, computer **152** may be a computer affiliated with an individual, such as an individual associated with acquiring bank **102a** or one of partners **116a**, **116b**, **116c**. As another example, computer **152** may be an intermediary computer between another user device (not shown), such as a tablet, smartphone, or other device, and processor **150**. For example, computer **152** may be configured to receive data from a tablet, smart phone, or other user device and transmit received data to processor **150**, and/or receive data from processor **150** and send received data to a tablet, smart phone, or other device. In some embodiments, computer **152** may include a display screen, such as a personal computer monitor, tablet screen, or mobile phone screen, on which a graphical user interface may be displayed.

(33) Database **154** may be a computer database configured to store digital data. Database **154** may be located on, for example, one or more hard drives, such as SATA drives or solid state drives. Database **154** may be configured to send and/or receive data to and/or from processor **150** and matching engine **156**. For example, processor **150** may receive data regarding one or more of new leads **114a**, **114b**, and may transmit such data to database **154** for storage. In some embodiments, database **154** may be configured to respond to queries for data from processor **150** and/or matching engine **156**. For example, matching engine **156** may send a query to database **154** for all data relating to, for example, one or more of partners **116a**, **116b**, and **116c**. Database **154** may respond to such a query by transmitting the requested data to matching engine **156**.

(34) Matching engine **156** may be a computer, including a computer processor, configured to execute one or more narrowing algorithms on data received from processor **150** and database **154**, such as, for example, the narrowing algorithms depicted in FIG. **12**, in order to generate a list of potential partners to match with a new lead. For example, matching engine **156** may be configured to receive one or more parameters from processor **150** including one or more narrowing attributes relating to a new lead **114a**, submit one or more queries for data regarding one or more partners such as partners **116a**, **116b**, **116c** from database **154**, receive data from database **154** in response to the one or more queries, and execute an algorithm based on the one or more received queries from processor **150** in order to narrow the data received from database **154**. Algorithms that may be performed by matching engine **156** are described in further detail below.

(35) In some embodiments, processor **150** may be a part of matching engine **156**. In further alternative embodiments, processor **150**, database **154**, and matching engine **156** may all be part of a single integrated computer system.

(36) FIG. **2** depicts an exemplary view of a graphical user interface showing an embodiment of a dashboard **200** for enabling acquiring bank **102a** to evaluate partners such as **116a**, **116b**, **116c** and match partners such as partners **116a**, **116b**, **116c** with one or more new leads, such as new leads **114a**, **114b**. Dashboard **200** may be an interactive interface tailored to show a summary of information relevant to payments processing network **100**, and specifically relevant to the account of a particular user **210**. For example, dashboard **200** may show a referrals module **202**, a purchased leads module **204**, a web inquiries module **206**, and a quotes module **208** for a given administrative user **210** associated with acquiring bank **102a**, wherein each of the modules **202**, **204**, **206** and **208** are customized to show information particular to acquiring bank **102a**. Dashboard **200** may also show, for example, modules such as an industry summary module **214** and a lead history chart **212**. The graphical user interface may be generated by, for example, one or more processors, and may be displayed on one or more local or remote monitors, such as computer

monitors, tablet screens, and/or mobile phone screens.

(37) Dashboard **200**, and each module displayed on dashboard **200**, may be customizable by dashboard user **210** or by, for example, a back-end dashboard manager, such as a web administrator. Dashboard **200** may also be automatically configured by a processor based on known permissions specific to user **210**. Dashboard **200** may present a variety of optional modules for display based on the identity of user **210**. Multiple different dashboards, each suited to a different user, may automatically be configured and/or generated in parallel.

(38) Data displayed by dashboard **200** (and other embodiments of graphical user interfaces described herein) may be received from, for example, a database containing data about partners **116a**, **116b**, **116c**, merchants **112a**, **112b**, new merchant leads **114a**, **114b**, acquiring bank **102a**, payment networks **104a**, **104b**, and/or issuing banks **106a**, **106b**. Such data may generally include biographical information, such as company names, addresses, contact information, etc., as well as performance history, statistics, and other information relating to the participation of each entity in payments processing network **100**.

(39) User **210** depicted in FIG. 2 may be, for example, affiliated with acquiring bank **102a**. Alternately, user **210** may be any individual affiliated with an entity in payments processing network **100**, such as partner **116a**, merchant **112a**, new merchant lead **114a**, acquiring bank **102a**, or issuing bank **106a**. For user **210** employed by acquiring bank **102a**, referrals module **202** may summarize how many referrals have been made to acquiring bank **102a** from a given referral service. User **210** may require an account designed to allow user **210** to sign into the dashboard interface.

(40) For user **210** affiliated with an acquiring bank **102a**, the purchased leads module **204** depicted in FIG. 2 may reflect a number of contracts with merchants **112a**, **112b** and/or new merchant leads **114a**, **114b** purchased by acquiring bank **102a** from another entity, such as a vendor, contractor, advertiser, or any other acquiring bank, such as acquiring bank **102b**. The web inquiries module **206** may indicate, for example, the number of web inquiries acquiring bank **102a** has received at its website from new merchant leads such as new merchant leads **114a**, **114b** in a given period of time. The quotes module **208** may reflect overall amounts quoted to potential new merchant leads **114a**, **114b**.

(41) With respect to a user **210** affiliated with, for example, partner **116a** or an acquiring bank **102a**, the industry summary module **214** may show, for example, the industries in which user **210** and/or partner **116a** has clients. For example, partner **116a** may work with leads **112a**, **112b** in a variety of industries, which may be displayed in the industry summary module **214** as, for example, a pie chart or a list. Also, with respect to user **210** affiliated with partner **116a**, the lead history chart **212** may show, for example, a history of leads generated and/or assigned by partner **116a** and/or user **210** over a given period of time.

(42) While a variety of modules are depicted in dashboard **200** pictured in FIG. 2, it is to be understood that the modules and their content may vary depending on user **210**, the affiliation of user **210** (e.g., with partner **116a**, lead **112a** or **112b**, new lead **114a** or **114b**, acquiring bank **102a**, or issuing bank **106a** or **106b**), the permissions granted to user **210**, and other variables.

(43) FIG. 3 depicts an exemplary view of a partner management user interface **300** providing partner management tools for, e.g., managing partner information, partner status, partner interactions, etc. Partner management user interface **300** may, for example, be accessible via a link from dashboard **200**. Partner management user interface **300** may also be tailored to user **210**, such as a user affiliated with acquiring bank **102a** or merchant **112a**. Partner management user interface **300** may display, for example, a partner list **302**, which may display a list of partners (e.g., partners **116a**, **116b**, **116c**) by name, contact, and/or status. For example, the partners displayed in the partner list **302** may be all or some of the partners affiliated with the organization of user **210** (e.g., acquiring bank **102a**). Each partner in the partner list **302** may be displayed along with, for example, an action selection menu **308**, which may allow for the selection of one or more actions

relating to the partner. For example, the action selection menu **308** may include an option to edit the partner entry in the partner management user interface, an option to send the listed partner an activation link which the partner may use to change its status, and/or a manual activation option, which may allow the user to manually change the status of the listed partner. The status of the partner may be based, for example, on whether the partner is currently accepting or recruiting new leads (e.g., new leads **114a**, **114b**) as clients and/or for participation in a partnership with acquiring bank **102a**. Partner management user interface **300** may provide options for viewing partners having a particular status; for example, partner management user interface **300** may have an active/inactive tab **304** which may display partners having an active and/or inactive status, and a pending activation tab **306**, which may display partners having a “pending activation” status. As with dashboard **200**, the information displayed on partner management user interface **300**, such as the listing of partners and/or the permissions to edit partner entry information on the partner management user interface **300** may vary with the identity of user **210** for which partner management user interface **300** is generated.

(44) FIG. 4 depicts an exemplary view of a partner details user interface **400** for viewing and managing details about a partner, such as partners **116a**, **116b**, **116c**. Partner details user interface **400** may, for example, be accessible by clicking on the name of a partner in partner management user interface **300**. Partner details user interface **400** may include a plurality of selectable tabs, including a partner account profile tab **402**, a service region tab **404**, an industry coverage tab **406**, a user settings tab **408**, and a partner program tab **410**.

(45) Partner account profile tab **402** may include details associated with a partner, such as the partner name, main contact name, phone number, fax number, address, website, etc. Partner account profile tab **402** may also show a sales method **403**, which may be the method by which the partner associated with the partner account accomplishes sales. For example, partner **116a** may accomplish outside sales, in which case sales method **403** will reflect that partner **116a** accomplishes outside sales. Partner account profile tab **402** may also list the number of sales staff associated with the partner account. Partner account profile tab **402** may also list the number of monthly locations at which the partner appears if, for example, the partner makes periodic appearances at a variety of locations. Alternately, partner account profile tab **402** may specify that the partner has a permanent business location. Partner account profile tab **402** may provide further details about the partner, such as whether the partner supports multiple locations, whether the partner supports deployment to a remote lead, whether the partner participates in optional programs, and whether the partner should be excluded from a lead pairing process. Depending on the permissions of user **210**, the information viewable in partner account profile tab **402** may be editable by user **210**.

(46) FIG. 5 depicts another exemplary view of a partner details user interface **400**, displaying industry coverage tab **406**. Upon being selected, industry coverage tab **406** may display a plurality of industry categories **502**, which may each contain a plurality of selectable industry subcategories corresponding to industries in which one or more merchants (e.g., merchants **112a**, **112b**) or potential new leads (e.g., new merchant leads **114a**, **114b**) operate. If a subcategory is selected using, for example, the check box next to the subcategory name, that may indicate that the partner is or has been paired with a merchant in that industry subcategory, and/or that the partner has particular capabilities or specialties relating to that industry subcategory. Depending on the permissions of user **210**, the information viewable in partner account profile tab **402** may be editable by user **210**.

(47) FIG. 6 depicts another exemplary view of partner details user interface **400**, displaying partner program tab **410**. Upon being selected, and for some users **210** (such as, for example, users affiliated with acquiring bank **102a**), partner program tab **410** may display modules relating to the process of pairing partners such as partners **116a**, **116b**, **116c** with new leads such as new leads **114a**, **114b**. For example, partner program tab **410** may include a lead sharing module **602**, which

may include an option to enable the lead sharing module and an option to set the default status for pairing a partner, such as partner **116c** with a new lead, such as new lead **114a** (e.g., a default status of “no” indicates that the partner **116c** is not to be included as an option for pairing with new lead **114a**, by default). Partner program tab **410** may also include, for example, a promotions module **604**, which may include an option for specifying that sales staff of the partner participate in one or more promotion programs, and a promotions channel through which promotions programs may be managed for the partner.

(48) FIG. 7 depicts an exemplary view of a lead management user interface **700** for depicting a plurality of new leads (e.g., new leads **114a**, **114b**) and/or merchants (e.g., merchants **112a**, **112b**), and information about them. Lead management user interface **700** may be accessible, for example, via a link from dashboard **200**. Lead management user interface **700** may include a plurality of tabs including, for example, a new leads tab **702**, a “my leads” tab **704**, an “all leads” tab **706** and a search function **708**. The new leads tab **702** may include a list of new leads **710**, which may include details about each new lead such as the date the lead was entered into the list, the lead source, the contact person at the lead, the company name, and the location of the lead. The list **710** may further include a “manage” link for each new lead in the list which, upon selection, may allow for the entry or editing of details for each new lead.

(49) As with the prior graphical user interfaces discussed above, lead management user interface **700** may show more or less content depending on the identity and/or affiliations of user **210** for which lead management user interface **700** is generated, and may allow or disallow editing of the content displayed on lead management user interface **700** depending on permissions granted to user **210**. For example, a lead management user interface designed for a user employed by partner **116a** may display only a list of new leads **710** which may potentially be paired with partner **116a**. Alternately, a lead management user interface designed for a user employed by a partner **116a** may not display a new leads tab **702** or an “all leads” tab **706**, and may instead only show the “my leads” tab **704**.

(50) FIG. 8 depicts an exemplary view of a lead details user interface **800**, showing the details pertaining to a given merchant (e.g., merchant **112a**, **112b**) or new lead (e.g., new lead **114a**, **114b**). Lead details user interface **800** may be accessible, for example, via a link from a lead entry in the new leads list **710**. Lead details user interface **800** may include a lead biography module **802**, an interactions module **806**, and an assigned partners module **810**.

(51) Lead biography module **810** may contain, for example, identification details for the merchant or new lead of interest including, for example, the date the merchant or new lead was entered, its company name, the name of its contact person, the address, email, and website address of the merchant or new lead. Lead biography module **810** may also include a message link, which may lead to an electronic form in which a message can be typed to the merchant, new lead, or a contact person associated with the merchant or new lead. The interactions module **806** may include a set of markers to indicate whether the merchant or new lead has been called, emailed, or visited, whether the merchant or new lead has received a demonstration, whether the merchant or new lead is a multi-location opportunity, and whether the lead systems have been synchronized to services provided by a partner. Interactions module **806** may also include a private comments section **808**, in which comments regarding interactions with the merchant or new lead can be recorded. Assigned partners module **810** may include several fields in which partners assigned thus far to the merchant or new lead may be specified. Assigned partner module **810** may further include a comment section **812**, in which comments about partner assignments may be kept.

(52) FIG. 9 depicts an alternate version of a lead management user interface **900** for managing details of a merchant (e.g., merchant **112a**, **112b**) or new lead (e.g., new lead **114a**, **114b**). In this embodiment, a lead details tab **902** may be another tab adjacent to tabs similar to those that were depicted in lead management user interface **700**. Lead details tab **902** may include a lead biography module **912** listing, for example, the date, name, contact name, address, phone number, and email

address of the merchant or new lead. Lead details tab **902** may also include a lead interactions module **914**, having a set of markers similar to those in interactions model **806** represented in FIG. **8**.

(53) FIG. **10** depicts an embodiment in which lead management user interface **900** may further have a lead pairing tool tab **910**. In the case of a new lead (e.g., new lead **114a**, **114b**), the lead pairing tool tab **910** may include the display of a lead pairing questionnaire **1002**, which may contain a list of questions that have been, or will be, directed to a new lead via, for example, a new lead user interface (see FIG. **11**). Lead pairing tool tab **910** may also include a recommended providers list **1004**, which may display a list of selectable potential partners **116** that may be suitable for pairing with the new lead **114**, and which may change depending on one or more answers to questionnaire **1002** received from the new lead. Recommended providers list **1004** may include an assignment button **1006**. A partner detail window **1008** may be viewable upon mousing over or otherwise indicating a partner in the recommended providers list **1004**. Upon one or more potential partners being selected and the assignment button **1006** being pressed, the one or more selected potential partners may be paired with the new lead. Upon such a new pairing, the partner and/or the new lead may be sent contact information and other details to allow the partner to begin assisting the new lead with merchant boarding and other electronic payments processing-oriented services.

(54) FIG. **11** depicts a new lead user interface **1100** displaying questionnaire **1002**. New lead user interface **1100** may be viewable to a new lead **114** on an electronic screen, such as a computer monitor, mobile phone screen, or tablet screen. New lead user interface **1100** may be generated at a remote processor, such as a processor operated by an acquiring bank (e.g., acquiring bank **102a**), and may be transmitted to a new lead over an electronic network, such as the Internet. Questionnaire **1002** may be populated with a series of questions having a plurality of selectable answers, e.g., **1102**, **1106**, **1108**, **1110**, and/or fillable answer fields **1104**. Questionnaire **1002** may also have a submission button **1112**, which may submit questionnaire **1002** with the provided answers.

(55) Questionnaire **1002** may generally include any questions that may assist in automatically identifying a partner that is suited to being paired with the new lead. For example, the questions in questionnaire **1002** may be directed to biographical information about the new lead, such as the new lead's business type, location, number of locations, and age of business. The questions may also be directed to the new lead's method of doing business, such as whether the new lead has a technical person on staff to oversee the point of sale (POS), how the new lead plans on paying for a POS system, whether the new lead would like a quote on payment processing and gift cards, and whether the new lead prefers a mobile tablet POS as opposed to, for example, a desktop POS or POS with a wired connection.

(56) FIG. **12** depicts a schematic diagram of an algorithm **1200** executed by a matching engine, such as matching engine **156** shown in FIG. **1B**, to pair partners (e.g., partners **116a**, **116b**, **116c**) with new leads (e.g., new leads **114a**, **114b**). Algorithm **1200** may be used to narrow down a list of potential partners for pairing with one or more new leads, based on either answers to questionnaire **1002** received from the one or more new leads and/or based on other parameters that may render a potential partner suitable for pairing with a new lead. According to a first step **1204**, a list of potential partners **1202** may be received. The list **1202** may be based, for example, on eligible accounts. Eligible accounts may be the accounts of partners which are active (i.e., currently accepting pairings with new leads), and which have not been paired with a new lead in a specified period of time, such as an hour, day, etc. According to a second step **1206**, list **1202** may be narrowed by, for example, removal of partners that do not meet a certain threshold, such as a threshold of lead recruitment success or a threshold approval rating received from merchants (e.g., merchants **112a**, **112b**). According to a next step **1208**, list **1202** may be further narrowed by, for example, removal of partners that are not a geographical match with the new lead. According to a

next step **1210**, the list **1202** may be further narrowed by, for example, removal of partners that do not provide vertical coverage in the area of business of the new lead. For example, if the new lead is a bar, then partners that do not provide service to (or specialize in service to) food and beverage service business may be removed from list **1202**. According to a next step **1212**, list **1202** may be sorted in an order of preference by, for example, the percentage **1214** of new lead accounts closed or won by each partner in list **1202**.

(57) As will be apparent to one of ordinary skill in the art, the criteria by which list **1202** may be narrowed may occur in the above-described order, but may also occur in any other suitable order for identifying partners eligible to be paired with new leads. In some embodiments, some criteria may be prioritized over others, e.g., based on priorities expressed by a new lead, partner, or acquiring bank. For example, in some embodiments, a narrowing step **1210** by vertical coverage may be performed before the narrowing step **1208** by geographical match.

(58) Moreover, as will be apparent to one of ordinary skill in the art, the narrowing criteria depicted in FIG. **12** are not exhaustive. Other narrowing criteria may be applied to list **1202** of potential partners in addition to, or instead of, the criteria shown in FIG. **12**. For example, list **1202** may be narrowed by removing, e.g., partners that do not provide a given service that a new lead needs, such as tablet-based POS terminals instead of desktop-based POS terminals. Other examples of narrowing criteria, or criteria by which potential partners may be sorted, include the number of service staff available at potential partners, the relative physical distance between the new lead and the potential partners, and/or the current capacity for new leads that each potential partner has. Other narrowing criteria may be ascertained by those of ordinary skill in the art, and may be added to or substituted into algorithm **1200**.

(59) FIG. **13** depicts other aspects of a graphical user interface relating to services between partners, merchants and/or new leads, according to some embodiments of this disclosure. Specifically, FIG. **13** depicts a promotions detail interface **1300**, which may display a partner identity module **1302** and a points module **1304** for a partner identified in the partner module **1302**. The promotions detail interface may also show a variety of promotion performance modules, such as a card activation module **1306**, a gift sales module **1308**, a security promotions module **1310**, and a target/achieved leads module **1312**. The promotions detail interface **1300** may also display options for submitting promotions materials, such as a certification submission option **1314**, a promotional story submissions option **1316**, and a rebranding submission option **1318**. Promotions may be geared towards specific goals, such as increasing the participation of partners and merchants in enhanced payments processing security programs. Partners may earn points for completing, participating in, and/or encouraging merchants and/or new leads to participate in, various promotions which may be reflected on the promotions detail interface **1300**. Points earned may be stored in a database or server along with other information about partners, new leads, and other entities in payment processing network **100**. The points gained by partners may be a further criterion by which partners are sorted in algorithm **1200** depicted in FIG. **12**, or in the methods described in FIGS. **16** and/or **17** below.

(60) FIG. **14** depicts a graphical user interface showing a promotions leaderboard **1400**, which may show summaries of partner performance and point accumulation with regard to, for example, promotions, such as the promotions detailed in promotions detail interface **1300**. The promotions leaderboard may include, for example, an active partners summary **1402**, an average points summary **1404**, a top partners summary **1404**, and a security promotions summary **1408**. The promotions leaderboard **1400** may include a list of partners **1410**, which may be sortable by, for example, individual name, company name, and/or points balance.

(61) FIG. **15** shows further aspects of a graphical user interface according to some embodiments of this disclosure. A user settings interface **1500** is depicted, which may show a user list **1502** of users which may have access to one or more of the above-described graphical user interfaces.

(62) Depending on the account of user **210** viewing user settings interface **1500**, promotions

leaderboard **1400**, promotions detail interface **1300** and/or other embodiments of interfaces disclosed herein, the interfaces may allow more or less information to be shown. Further, and depending on the account and permissions of user **210**, the information for each user in the user list **1502** may be editable.

(63) FIGS. **16** and **17** relate generally to methods of generating and displaying a plurality of potential partners (such as partners **116a**, **116b**, **116c**) for pairing with new leads (such as new leads **114a**, **114b**) using, e.g., the embodiments of graphical user interfaces described above, such as (but not limited to) lead management user interfaces **700**, **900**, and new lead user interface **1100**.

(64) Specifically, FIG. **16** depicts a selection method **1600** of selecting and displaying a narrowed plurality of potential partners to pair with a new lead. According to step **1602**, an electronic database comprising a plurality of potential partners and a plurality of attribute categories may be received at a processor, each potential partner having an attribute corresponding to each attribute category. This received database may be, for example, a database from which data is taken to generate one or more user interfaces described above. The plurality of potential partners may be a plurality of partners including, for example, one or more of partners **116a**, **116b**, **116c**, such as a potential partner list **1202**, as depicted in FIG. **12**. The plurality of attribute categories may include categories of biographical and other information about the plurality of potential partners and/or merchants already paired with each of the plurality of potential partners, that may affect whether a potential partner should be paired with a new lead, such as partner status (active or inactive), partner location, capacity for new pairings, types of businesses served, company sizes of paired merchants, number of partner locations, number of partner staff members, prior success in completing merchant boarding with merchants, prior satisfaction ratings from merchants, and types of services provided by each of the plurality of partners (e.g., POS terminal servicing, gift card services, security services, etc.). Further attribute categories may cover any information that may be relevant to pairing each of the plurality of potential partners with one or more new leads.

(65) According to step **1604**, a plurality of threshold eligibility attributes may be received at the processor, each threshold eligibility attribute corresponding to at least one of the plurality of attribute categories. The plurality of threshold eligibility attributes may be attributes used to initially remove potential partners from the plurality of partners that may generally not be willing or able to pair with new leads. In some embodiments, the threshold eligibility attributes may be, for example, some or all of the partner criteria discussed with respect to algorithm **1200** depicted in FIG. **12**. In alternate embodiments, the threshold eligibility attributes may include information that affects the eligibility of partners independently of the identity of any new leads, such as partner capacity for being paired with new leads, prior success in completing merchant boarding with prior paired leads, i.e., merchants, and/or satisfaction ratings provided by prior paired merchants.

(66) According to step **1606**, the processor may be used to automatically narrow the plurality of potential partners using one or more of the plurality of threshold eligibility attributes as one or more narrowing attributes (according to, for example, the steps depicted in FIG. **17**). In some embodiments, the processor may first prioritize the plurality of threshold eligibility attributes in a particular order of priority, based on, for example, which threshold eligibility attributes are most important to a new lead. In such embodiments, the narrowing step **1606** may narrow the plurality of potential partners using the threshold eligibility attributes in their order of priority.

(67) According to step **1608**, the processor may be used to construct an electronic questionnaire, wherein the questionnaire comprises a plurality of interactive, displayable prompts to a graphical user interface. This questionnaire may be, for example, a questionnaire such as questionnaire **1002** depicted in FIGS. **10** and **11**. The questionnaire may be, for example, constructed of questions prompting a new lead (such as new leads **114a**, **114b**) to provide information useful to pairing the new lead with one or more partners (such as partners **116a**, **116b**, **116c**). In some embodiments, for example, the questionnaire may be pre-constructed prior to any information being received about a new lead. In alternate embodiments, the questionnaire may be constructed after information about a

new lead has been received, e.g., through a prior electronic or written form submitted by the new lead. In such embodiments, the questionnaire may be, for example, constructed of only questions to which a given database having data on the new lead does not have answers.

(68) According to step **1610**, the processor may be used to transmit the questionnaire to an interactive lead user interface over an electronic network. In some embodiments, the interactive lead user interface may be, for example, an interface such as lead user interface **1100** depicted in FIG. **11**. In alternate embodiments, the lead user interface may not be separate from a general dashboard **200** or lead management user interface **900**, depicted in FIG. **10**.

(69) According to step **1612**, one or more answers may be received from the lead user interface, over the electronic network, and at the processor, wherein each answer comprises at least one lead attribute that corresponds to one of the plurality of attribute categories. For example, a new lead may complete one or more answers to questionnaire **1002** on, e.g., lead user interface **1100**, and may submit the questionnaire by selecting submission button **1112**, at which point questionnaire **1002** and answers may be received by the processor. Lead attributes may include any biographical or other information about the new lead, as has been described previously herein. Such biographical or other information (such as location, age, size of company, etc.) may correspond to one or more of the plurality of attribute categories.

(70) According to step **1614**, the processor may be used to automatically construct and store an electronic lead profile using the at least one lead attribute, wherein the lead profile comprises a lead identity, the at least one lead attribute, and identification of the attribute category to which the at least one lead attribute corresponds. Such an electronic lead profile may be created and stored in a database, or may be displayed on, for example, a graphical user interface, such as lead details interface **800** depicted in FIG. **8**, lead management interface **900** depicted in FIG. **9**, or both. In some embodiments, if prior information has already been collected regarding a new lead, a partial lead profile may already exist in one or more electronic databases. In such embodiments, the partial lead profile may be supplemented using the at least one lead attribute received. In alternative embodiments, a full lead profile may already exist; in such embodiments, this step **1614** may optionally not be performed.

(71) According to step **1616**, the processor may be used to automatically narrow the plurality of potential partners using the at least one lead attribute as at least one narrowing attribute (according to, for example, the steps depicted in FIG. **17**). In embodiments where more than one lead attribute is used, the processor may optionally be used to sort the lead attributes in an order of importance, e.g., based on which lead attribute should be considered the highest priority in pairing a potential partner with a new lead. In such embodiments, the narrowing step **1616** may narrow the plurality of potential partners using the lead attributes in their order of importance.

(72) According to step **1618**, the processor may be used to automatically add the narrowed plurality of potential partners to the lead profile as, for example, a database entry or plurality of entries associated with the rest of the lead profile.

(73) According to step **1620**, the lead profile and narrowed plurality of potential partners may be displayed on a graphical user interface. For example, in some embodiments, the lead profile and narrowed plurality of potential partners may be displayed on lead management interface **900** under lead pairing tool tab **910**, as depicted in FIG. **10**. In some embodiments, the narrowed plurality of potential partners may be displayed in a selectable list, such that a user viewing the graphical user interface may select one or more partners and pair them with a new lead. In some embodiments, as depicted in FIG. **10**, partner detail window **1008** may be viewable for each of the potential partners in the narrowed plurality of potential partners, to potentially assist a user in making a final selection of a potential partner to pair with a new lead. In alternate embodiments, the processor may further automatically select one or more of the narrowed plurality of potential partners to be paired with a new lead. In such embodiments, the processor may automatically send a notification to the selected partner, the new lead, both, and/or a representative of either or both to notify them of the selection.

(74) FIG. 17 displays sub-steps in an automatic narrowing process **1700** that may be used to narrow a plurality of potential partners based on narrowing attributes. For example, narrowing process **1700** may be used to accomplish narrowing steps **1606**, **1616** (shown in FIG. 16) in method **1600**. According to a step **1702**, a plurality of narrowing attributes and a plurality of potential partners may be received at a processor. The plurality of narrowing attributes may be, for example, a plurality of threshold eligibility attributes received according to step **1604** in method **1600**, or may be lead attributes received according to step **1612** in method **1600**. In some embodiments, the plurality of narrowing attributes may be received in an order of priority. In alternate embodiments, the narrowing process **1700** may be done with a single narrowing attribute.

(75) Steps **1704** and onwards may be completed for each received narrowing attribute, and may be repeated for as many narrowing attributes as are received in the plurality of narrowing attributes. In embodiments in which the plurality of narrowing attributes have been received in an order of priority, step **1704** and onwards may be completed for each received narrowing attribute in the order of priority that the narrowing attributes were received in.

(76) According to a step **1704**, the processor may be used to automatically count the number of potential partners in the plurality of potential partners. According to a step **1706**, a processor may be used to determine whether the number of potential partners in the plurality of potential partners is greater than zero. If it is not, then according to step **1708**, the processor may be used to automatically display a notification on a graphical user interface. Such a notification may state, for example, that no potential partners are compatible with all of the narrowing attributes provided. In that case, because there are no potential partners to be sorted in the plurality of potential partners, the narrowing process **1700** may end here.

(77) In an alternate embodiment, the processor may be used to determine whether the number of potential partners in the plurality of potential partners is greater than one. In such an embodiment, and if only one potential partner remains in the plurality of potential partners, then as a part of displaying a notification according to step **1708**, the processor may identify the single potential partner in the plurality of potential partners, and indicate that the single potential partner is compatible with the most narrowing attributes out of the full plurality of potential partners. In further alternate embodiments, the processor may be used to determine whether the number of potential partners in the plurality of potential partners is greater than two, three, or any other number. In such alternate embodiments, as a part of displaying a notification according to step **1708**, the processor may identify the two, three, or other number of potential partners in the plurality of potential partners in the display.

(78) If, according to step **1706**, the number of potential partners in the plurality of potential partners is greater than zero, then according to a step **1710**, the processor may be used to automatically identify the potential partner attribute category to which the narrowing attribute corresponds. For example, the processor may be used to identify which attribute category in the plurality of attribute categories in the database received according to step **1602** the narrowing attribute belongs. For example, the processor may be used to identify whether the narrowing attribute is a geographic area, a type of business, or falls within some other category of information.

(79) According to a step **1712**, for each potential partner in the plurality of potential partners, a processor may be used to automatically identify the attribute of the potential partner corresponding to the identified attribute category in the electronic database, and automatically compare the identified attribute of the potential partner to the narrowing attribute. According to a step **1714**, the processor may be used to determine whether the identified attribute of the potential partner is compatible with the narrowing attribute. For example, if the identified attribute category is a geographic area, then the processor may be used to automatically determine what geographic areas are served by each potential partner, and may determine whether the narrowing attribute falls within the geographic area or areas served by each potential partner. As another example, if the identified attribute category is a threshold success percentage, then the processor may

automatically determine whether each potential partner has a success percentage higher than the threshold success percentage. If the identified attribute of the potential partner is compatible with the narrowing attribute, then the processor may allow the potential partner to remain in the plurality of potential partners. If not, then the processor may automatically eliminate the potential partner from the plurality of potential partners.

(80) While the above-referenced steps are described as being performed by one processor, the steps may be performed by multiple processors working independently or in a networked fashion, such as in series or in parallel. In embodiments in which multiple processors perform the above-described steps, the processors may be connected via wired or wireless connections over an electronic network, such as the Internet or a local area network connection.

(81) With regard to the graphical user interface embodiments and methods described above, the graphical user interface embodiments may be generated by one or more server processors, and may be displayed on one or more monitors or screens suitable for displaying user interfaces, such as computer monitors, tablet screens, and/or mobile phone screens. Optionally, data for generating the graphical user interface embodiments described herein may be generated or processed at one or more processors, and optionally transmitted over one or more electronic networks for rendering of user interfaces on one or more remote user processors and display on one or more remote monitors or screens.

(82) Any aspect set forth in any embodiment may be used with any other embodiment set forth herein. Every user interface set forth herein may be generated by any single computer processor and/or server or combination of computer servers and/or processors, and may be displayed locally or remotely on any suitable interactive display, monitor or screen. Each method set forth herein may also be performed by any single computer processor, server, and/or display, or any combination of computer servers, processors, and/or displays.

(83) It will be apparent to those skilled in the art that various modifications and variations can be made in the disclosed systems and processes without departing from the scope of the disclosure. Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the disclosure disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the disclosure being indicated by the following claims.

Claims

1. A method of automatically generating an electronic lead sharing display within a graphical user interface, the method comprising: storing, by a processor, an electronic lead profile in an electronic database using at least one lead attribute of a potential partner that corresponds to one of a plurality of threshold eligibility attributes; receiving, by the processor, a narrowing attribute; identifying, by the processor, an attribute category of the narrowing attribute; automatically identifying, by the processor, a respective attribute of a plurality of potential partners corresponding to the attribute category in the electronic database; automatically narrowing, by the processor, the plurality of potential partners using the respective attributes of the plurality of potential partners in an order of priority of the respective attributes, wherein a potential partner is eliminated from the plurality of potential partners if the respective attribute is not compatible with the narrowing attribute; automatically constructing, by the processor, a potential-partner-specific electronic questionnaire specific to each potential partner among the narrowed plurality of potential partners, wherein the questionnaire comprises a plurality of automatically determined answerable prompts displayed on a graphical user interface, the plurality of automatically determined answerable prompts being based on a plurality of threshold eligibility attributes of the potential partner; displaying, by the processor, the potential-partner-specific questionnaire in an interactive lead user interface; receiving, by the processor, one or more answers to the potential-partner-specific questionnaire via the interactive

- lead user interface over an electronic network, wherein each answer comprises the at least one lead attribute of the potential partner that corresponds to one of the plurality of threshold eligibility attributes; and automatically adding, by the processor, the narrowed plurality of potential partners to the electronic lead profile as a plurality of database entries associated with the electronic lead profile according to the at least one lead attribute.
2. The method of claim 1, further comprising: automatically comparing, by the processor, the narrowing attribute to the at least one lead attribute.
3. The method of claim 2, further comprising: automatically narrowing, by the processor, the plurality of potential partners based on a comparison of the narrowing attribute to the at least one lead attribute.
4. The method of claim 1, wherein the electronic lead profile comprises a lead identity, the at least one lead attribute, and identification of the attribute category to which the at least one lead attribute corresponds.
5. The method of claim 4, further comprising: displaying, by the processor, the electronic lead profile on a results user interface of the graphical user interface.
6. A device for automatically generating an electronic lead sharing display within a graphical user interface, the device comprising: a memory configured to store instructions; and a processor configured to execute the instructions to perform operations comprising: storing an electronic lead profile in an electronic database using at least one lead attribute of a potential partner that corresponds to one of a plurality of threshold eligibility attributes; receiving a narrowing attribute; identifying an attribute category of the narrowing attribute; automatically identifying a respective attribute of a plurality of potential partners corresponding to the attribute category in the electronic database; automatically narrowing the plurality of potential partners using respective attributes of the plurality of potential partners in an order of priority of the respective attributes, wherein a potential partner is eliminated from the plurality of potential partners if the respective attribute is not compatible with the narrowing attribute; automatically constructing a potential-partner-specific electronic questionnaire specific to each potential partner among the narrowed plurality of potential partners, wherein the questionnaire comprises a plurality of automatically determined answerable prompts displayed on a graphical user interface, the plurality of automatically determined answerable prompts being based on a plurality of threshold eligibility attributes of the potential partner; displaying the potential-partner-specific questionnaire in an interactive lead user interface; receiving one or more answers to the potential-partner-specific questionnaire via the interactive lead user interface over an electronic network, wherein each answer comprises the at least one lead attribute of the potential partner that corresponds to one of the plurality of threshold eligibility attributes; and automatically adding the narrowed plurality of potential partners to the electronic lead profile as a plurality of database entries associated with the electronic lead profile according to the at least one lead attribute.
7. The device of claim 6, wherein the operations further comprise: automatically comparing the narrowing attribute to the at least one lead attribute.
8. The device of claim 7, wherein the operations further comprise: automatically narrowing the plurality of potential partners based on a comparison of the narrowing attribute to the at least one lead attribute.
9. The device of claim 6, wherein the electronic lead profile comprises a lead identity, the at least one lead attribute, and identification of the attribute category to which the at least one lead attribute corresponds.
10. The device of claim 9, wherein the operations further comprise: displaying the electronic lead profile on a results user interface of the graphical user interface.
11. A non-transitory computer-readable medium storing instructions that, when executed by a processor of a device, cause the processor to perform operations comprising: storing an electronic lead profile in an electronic database using at least one lead attribute of a potential partner that

corresponds to one of a plurality of threshold eligibility attributes; receiving a narrowing attribute; identifying an attribute category of the narrowing attribute; automatically identifying a respective attribute of a plurality of potential partners corresponding to the attribute category in the electronic database; automatically narrowing the plurality of potential partners using the respective attributes of the plurality of potential partners in an order of priority of the respective attributes, wherein a potential partner is eliminated from the plurality of potential partners if the respective attribute is not compatible with the narrowing attribute; automatically constructing a potential-partner-specific electronic questionnaire specific to each potential partner among the narrowed plurality of potential partners, wherein the questionnaire comprises a plurality of automatically determined answerable prompts displayed on a graphical user interface, the plurality of automatically determined answerable prompts being based on a plurality of threshold eligibility attributes of the potential partner; displaying the potential-partner-specific questionnaire in an interactive lead user interface; receiving one or more answers to the potential-partner-specific questionnaire via the interactive lead user interface over an electronic network, wherein each answer comprises the at least one lead attribute of the potential partner that corresponds to one of the plurality of threshold eligibility attributes; and automatically adding the narrowed plurality of potential partners to the electronic lead profile as a plurality of database entries associated with the electronic lead profile according to the at least one lead attribute.

12. The non-transitory computer-readable medium of claim 11, wherein the operations further comprise: automatically comparing the narrowing attribute to the at least one lead attribute.

13. The non-transitory computer-readable medium of claim 12, wherein the operations further comprise: automatically narrowing the plurality of potential partners based on a comparison of the narrowing attribute to the at least one lead attribute.

14. The non-transitory computer-readable medium of claim 13, wherein the operations further comprise: displaying the electronic lead profile on a results user interface of the graphical user interface.

15. The method of claim 1, further comprising: automatically updating, by the processor, the graphical user interface with at least one selectable potential partner element based on the one or more answers to the potential-partner-specific questionnaire received via the interactive lead user interface.

16. The method of claim 1, further comprising, for the narrowing attribute, using the processor to automatically count a number of potential partners in the plurality of potential partners, and, if the number of potential partners is no greater than a set number, then using the processor to automatically display a notification on the graphical user interface.

17. The device of claim 6, the operations further comprising: automatically updating, by the processor, the graphical user interface with at least one selectable potential partner element based on the one or more answers to the potential-partner-specific questionnaire received via the interactive lead user interface.

18. The device of claim 6, the operations further comprising, for the narrowing attribute, using the processor to automatically count a number of potential partners in the plurality of potential partners, and, if the number of potential partners is no greater than a set number, then using the processor to automatically display a notification on the graphical user interface.

19. The non-transitory computer-readable medium of claim 11, the operations further comprising: automatically updating, by the processor, the graphical user interface with at least one selectable potential partner element based on the one or more answers to the potential-partner-specific questionnaire received via the interactive lead user interface.

20. The non-transitory computer-readable medium of claim 11, the operations further comprising, for the narrowing attribute, using the processor to automatically count a number of potential partners in the plurality of potential partners, and, if the number of potential partners is no greater

than a set number, then using the processor to automatically display a notification on the graphical user interface.
